

1 **Supplementary Information**

2 **Manuscript title:** Focused-Universe Reanalysis of Complement, Coagulation, and Extracellular-
3 Matrix Transcriptional Programs in Diabetic Kidney Disease

4 **Authors:** AUTHOR LIST REQUIRED BEFORE SUBMISSION

5 **Supplementary Figure S1**

6 Complete three-study REML estimates with modified Hartung–Knapp 95% confidence intervals
7 for the 125 genes with complete mapping.

8 **Supplementary Table S1**

9 Complete 145-gene random-effects results, including REML tau-squared, unmodified and
10 modified Hartung–Knapp uncertainty, prediction intervals, and 145-family FDR.

11 **Supplementary Table S2**

12 External GSE30528 gene-level results for all 145 genes, including mapping status, discovery
13 direction, external effect, Welch P value, and 145-family FDR.

14 **Supplementary Table S3**

15 Nine-pathway sample-label permutation results. All genes were recomputed jointly in each of
16 10,000 label permutations.

17 **Supplementary Table S4**

18 Per-study Hedges' g estimates for GSE142025 C1, GSE96804, GSE30528, and secondary same-
19 study compartment GSE30529.

20 **Supplementary Table S5**

21 Complete 145-row focused-universe lineage table, including the prior outcome-dependent
22 prioritization status. This table is supplied for transparency and is not an inferential result.

23 **Supplementary Figure S2**

24 Unsupervised PCA diagnostics for GSE142025 C1, GSE96804, and GSE30528 using the 1,000
25 most variable genes within each dataset. These plots do not identify batch effects in the absence
26 of batch labels.

27 **Supplementary Table S6**

28 Sample-metadata availability audit for batch, age, sex, renal-function, proteinuria, albuminuria,
29 and medication fields.

30 **Supplementary Table S7**

31 GSE96804 sex-adjustment sensitivity for the complete 145-gene family.

32 **Supplementary Table S8**

33 PCA variance and disease-group diagnostic summaries for the three primary datasets.

34 **Supplementary Table S9**

35 Donor-disjoint GSE30529 gene-level sensitivity for the complete 145-gene family after removal
36 of nine inferred shared-donor samples.

37 **Supplementary Table S10**

38 Donor-disjoint GSE30529 pathway sample-label permutation results using a fixed 5-DKD versus
39 8-control allocation.

40 **Supplementary Table S11**

41 Gene-level significance-switch summary comparing GSE30528 with donor-disjoint GSE30529.
42 The display_color_code field provides the requested red/blue/purple/gray quick index without
43 embedding spreadsheet-only formatting.

44 **Supplementary Table S12**

45 Descriptive WGCNA integration for all 145 focused genes. WGCNA is internal to GSE96804
46 and is not external validation.

47 **Supplementary Table S13**

48 Gene-wise expression correlations across nine title-inferred cross-compartment donor pairs.
49 These correlations do not estimate covariance between cohort-level disease effects.

50 **Supplementary Table S14**

51 Complete sample-level metadata and analysis-role table for the primary datasets and the
52 GSE30529 sensitivity.

53 **Supplementary Figure S3**

54 Within-dataset sample-correlation heatmaps based on the 1,000 most variable genes.

55 **Supplementary Table S15**

56 Sample-level median-correlation diagnostics and predefined robust-z outlier flags.

57 **Supplementary Table S16**

58 Complete 145-gene primary meta-analysis sensitivity after excluding the flagged samples.

59 **Supplementary Table S17**

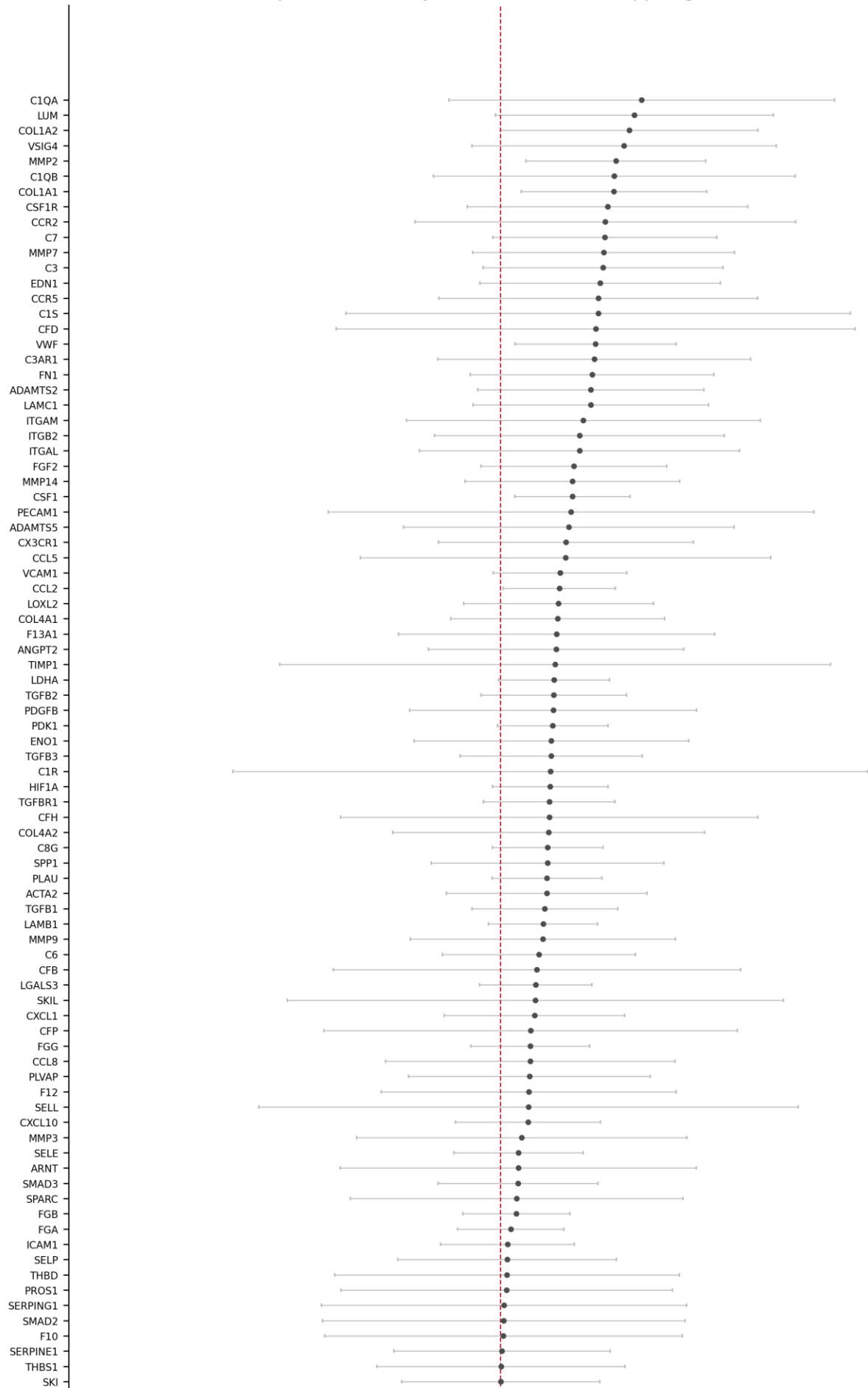
60 Source data for the Figure 2 fixed-gene forest display, including each study estimate, confidence
61 interval, random-effects weight, and pooled modified Hartung–Knapp interval.

62 **Supplementary Table S18**

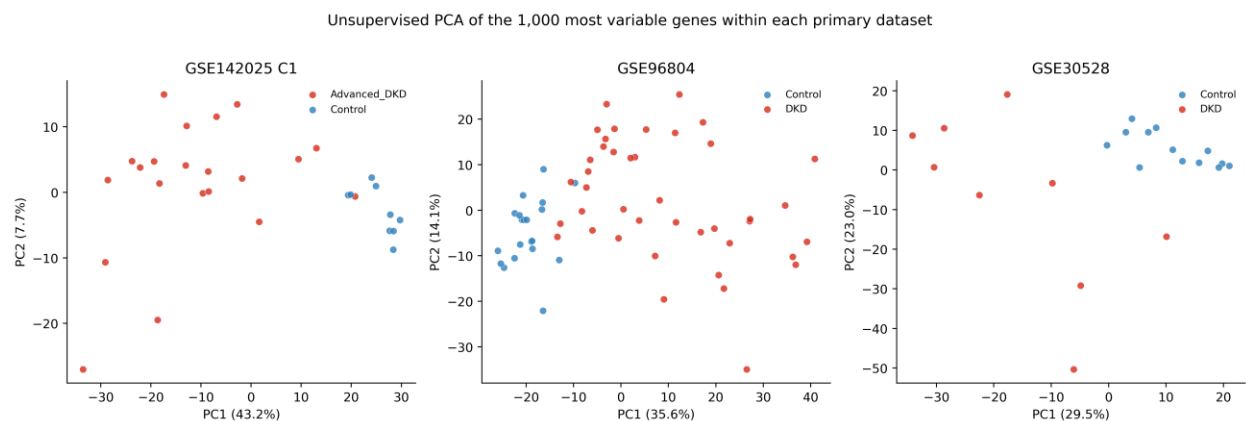
63 Pairwise within-dataset sample correlations underlying Supplementary Figure S3.

64 **Supplementary Figure S1 image**

Complete three-study estimates for 125 mapped genes

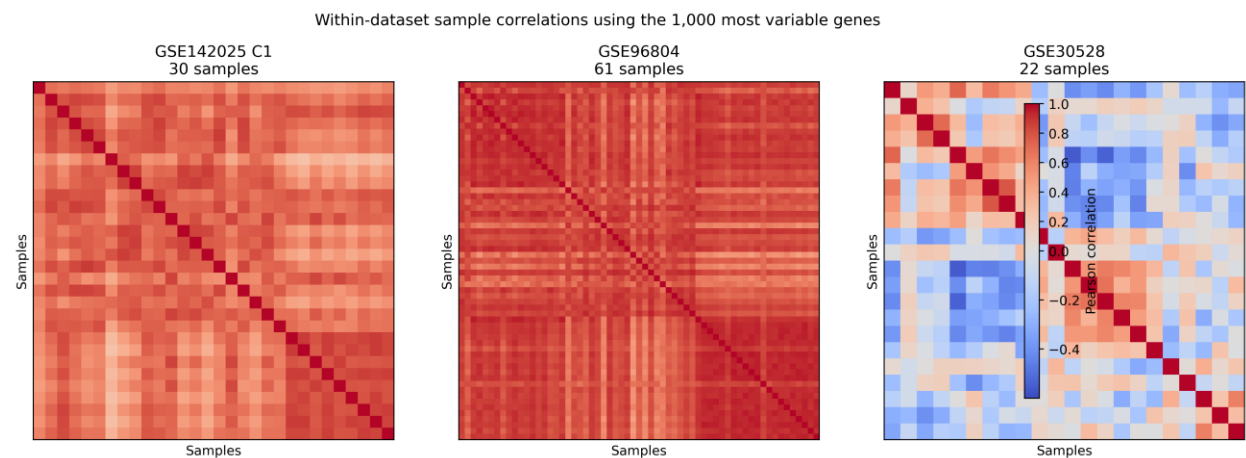


66 **Supplementary Figure S2 image**



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68 **Supplementary Figure S3 image**



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